BayesBall : A Comprehensive Framework to Predicting UCL Injury

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Outline

What is UCL Injury?

- UCL Injury, or Ulnar Collateral Ligament Injury, is the formal name for the tearing or stretching of the band of tissue on the inner side of the elbow.
- Better known as *Tommy John*, named after the first baseball player to have his elbow reconstructed in 1974.

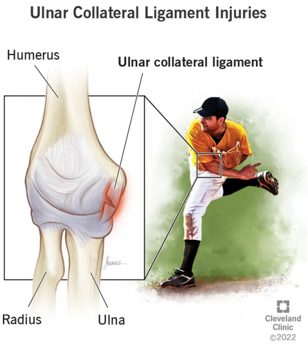

Figure 1: UCL Injury

Prior Research

- Tommy John Surgery is generally common, with 25 percent of MLB pitchers and 15 percent of Minor League baseball players having undergone the surgery. [2]
- Studies have also shown that pitch type (fastball, slider, etc) can have possible significance of risk of injury [1] and that other pitch mechanics (release position/ball spin) can also have relation to possible injury.

Research Framework

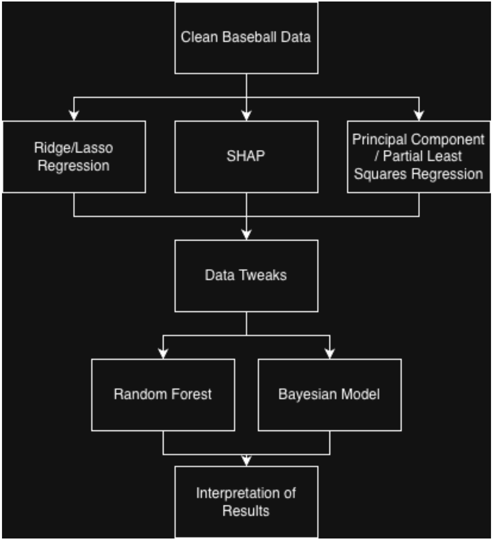

Figure 2: Current Research Framework

Data Collection

- Study used two cohorts, pitchers who underwent Tommy John Surgery and healthy control pitchers (non-UCL injured) from 2020–2024.
- Injured pitchers were sourced from a public surgery database, with two years of pre-surgery pitching data scraped for each player.
- Non-injured pitchers were pulled from a active list of MLB pitches between 2020-2024.
- All pitching data was collected using the `baseballR` package in R.

Data Description

- A total of 181,377 pitches where collected, with 133,878 of those used in the research.
- Pitches were classified using a binary operator depending if it came from a pitcher who had Tommy John Surgery.

Pitch Types		
Surgery	**0** **N = 96810**	**1** **N = 37068**
Changeup	11271	3588
Curveball	7934	1946
Fastball	35202	17538
Splitter	2070	903
Knuckle-Curveball	2439	1501
Sinker	17044	2577
Slider	16492	8581
Sweeper	4358	434

Figure 3: Pitch Counts

Variable Shrinkage and Selection

- A major goal of the first phase was variable selection. To approach this goal a variety of ML models were used to better understand the data and its variables.
- Lasso and Ridge Regression were used for Variable Shrinkage
- Principle Component Regression, Partial Least Squares Regression, and SHAP were used for better variable selection

Ridge Regression

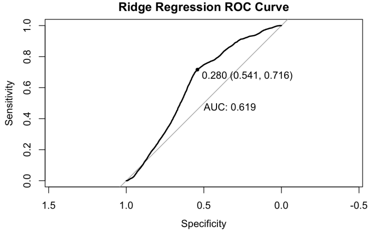

Figure 4: ROC Curve

	Truth	
Prediction	0	1
0	10495	2022
1	8906	5353

Figure 5: Confusion Matrix

Ridge Regression Output

- Accuracy - 0.5919
- Sensitivity - 0.7258
- Specificity - 0.5410

SHapley Additive exPlanations (SHAP)

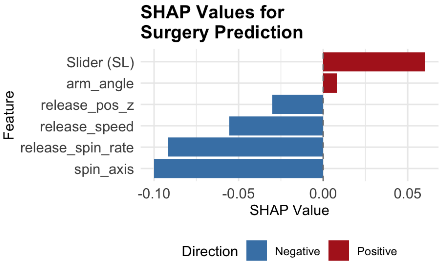

Figure 6: SHAP Values

- Multicollinearity is a major issue with the data, due to "bleeding" between variables and high interaction between pitches
- SHAP assists in separating terms, and finding predictive values Independently from each other.

Data Tweaks

- Weights from variable shrinkage and selection determined variables that moved on to be used in the final model creation.
- The majority of dropped variables suffered from limited instances of occurrence and had low sample populations for use.

Random Forest

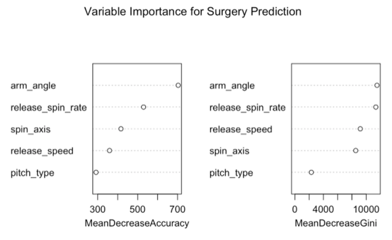

Figure 7: Random Forest Importance

	Truth	
Prediction	0	1
0	17618	2799
1	1744	4614

Figure 8: Confusion Matrix

- Random Forest was used since it produces deep trees that will provide a more accurate prediction of surgery.
- Significant improvement was seen in the prediction of surgery outcome by our variables now that specific information was dropped.

Random Forest Output

- Accuracy — 0.8303
- Sensitivity — 0.9099
- Specificity — 0.6224

Bayesian Regression Model using Stan (BRMS)

Likelihood:

$$y_{surgery} \sim \text{Bernoulli}(p)$$

Linear Predictor:

$$\begin{aligned}\hat{y}_{surgery} = b_0 &+ b_1 x_{release\_speed} + b_2 x_{arm\_angle} + b_3 x_{spin\_axis} \\ &+ b_4 x_{release\_spin\_rate} + b_5 x_{pitch\_type} + b_6 x_{release\_pos\_z} \\ &+ b_7 (x_{pitch\_FF} \times x_{pitch\_SL}) + b_8 (x_{pitch\_FF} \times x_{pitch\_FS}) \\ &+ b_9 (x_{pitch\_FF} \times x_{pitch\_KC})\end{aligned}$$

Priors:

$$b_j \sim \mathcal{N}(\beta_j, \sigma^2), \quad j = 0, 1, \ldots, 9$$

BRMS Outcomes

```
 Links: mu = logit
Formula: surgery ~ release_speed + arm_angle + spin_axis + release_spin_rate + pitch_type +
release_pos_z + pitch_FF:pitch_SL + pitch_FF:pitch_FS + pitch_FF:pitch_KC
   Data: data_complete_na (Number of observations: 133878)
  Draws: 5 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 5000

Regression Coefficients:
                  Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept             0.00      0.20    -0.39     0.40 1.00     6208     4676
release_speed        -0.02      0.00    -0.02    -0.01 1.00     5955     3882
arm_angle             0.02      0.00     0.02     0.02 1.00     6892     3826
spin_axis            -0.00      0.00    -0.00    -0.00 1.00     4596     3498
release_spin_rate     0.00      0.00     0.00     0.00 1.00     5434     4590
pitch_typeCU         -0.69      0.04    -0.77    -0.61 1.00     5140     3631
pitch_typeFF          0.42      0.03     0.37     0.49 1.00     4658     3611
pitch_typeFS          0.42      0.05     0.33     0.50 1.00     6631     3979
pitch_typeKC          0.33      0.04     0.23     0.41 1.00     5467     3575
pitch_typeSI         -0.61      0.03    -0.68    -0.55 1.00     5038     4085
pitch_typeSL          0.29      0.03     0.24     0.35 1.00     4608     3582
pitch_typeST         -1.32      0.06    -1.44    -1.21 1.00     5739     3890
release_pos_z        -0.16      0.02    -0.20    -0.12 1.00     8736     3452
pitch_FF:pitch_SL   -57.07    113.18  -297.27   176.64 1.95        7       18
pitch_FF:pitch_FS   -29.28     79.44  -232.48    80.37 1.56        9       23
pitch_FF:pitch_KC  -116.59    374.64 -1414.37   141.35 2.27        8       14
```

Figure 9: BRM Model Output

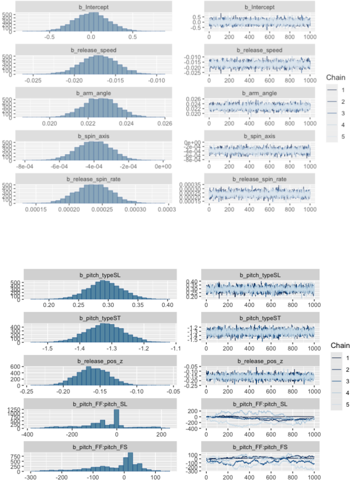

Figure 10: BRM Trace Plots

Future Research

- Focus on the interaction of pitch types and how pitch "groupings" effect surgical outcomes.
- Add additional information about pitchers into the models, taking into account age and other factors to better results.
- Publish current research into a academic journal.

Thank You

Any Questions?

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